

CO4 - AT2

Articulation

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Name :- Cloud Computing

1. Retail Data Optimization Problem

A retail company has physical stores and an online platform. It needs :- Real time inventory tracking

Demand forecasting

Personalized marketing recommendations

Solution :-

A Big Data Architecture can collect data from stores, websites, mobile applications, inventory systems, and Customer interactions.

Physical Address Stores
online platform
mobile app

↓
Data Ingestion

↓
Stream + Batch Processing

↓
Big Data Storage

↓
Real time stock

↓
Inventory management

↓
Demand forecasting

↓
Customer Analytics

↓
Marketing Recommendation

Technologies ÷

Kafka ÷ collect realtime sales and inventory events

cloud storage ÷ storage → stores large volume of data

Spark ÷ Perform data processing and Product data

NoSQL Database ÷ store flexible customer and Product data

Analytics models ÷

Demand forecasting ÷ Historical sales, seasonal trends, and customer behavior can be analyzed to predict future demand

Recommendation models ÷ Customer browsing, purchasing, and product interaction.

Benefits ÷ Real time inventory visibility, Better demand Prediction, Reducing stock-out, overstocking etc.

2. Social Media Fraud Detection Problem

A social media platform processes billions of posts, comments, likes, and shares. It needs to identify fake accounts and misinformation quickly.

Solution ÷

Use a real-time Big data streaming architecture.

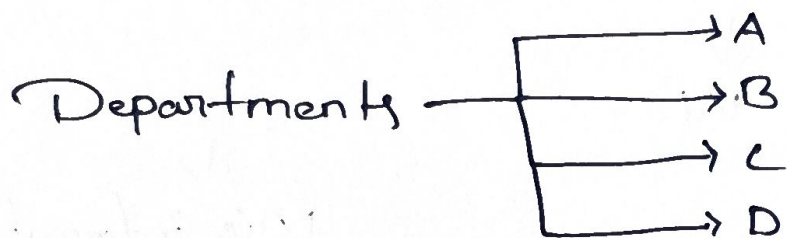
3. Government Data platform Problem

A government agency combines census, economic, and geospatial data from different departments. The system requires interoperability, Privacy and Security.

Solution:

Implement a centralized data governance and management framework.

Architecture:



↓
Data Ingestion

Data integration

Government Big Data Platform Processing

Data Governance

Privacy

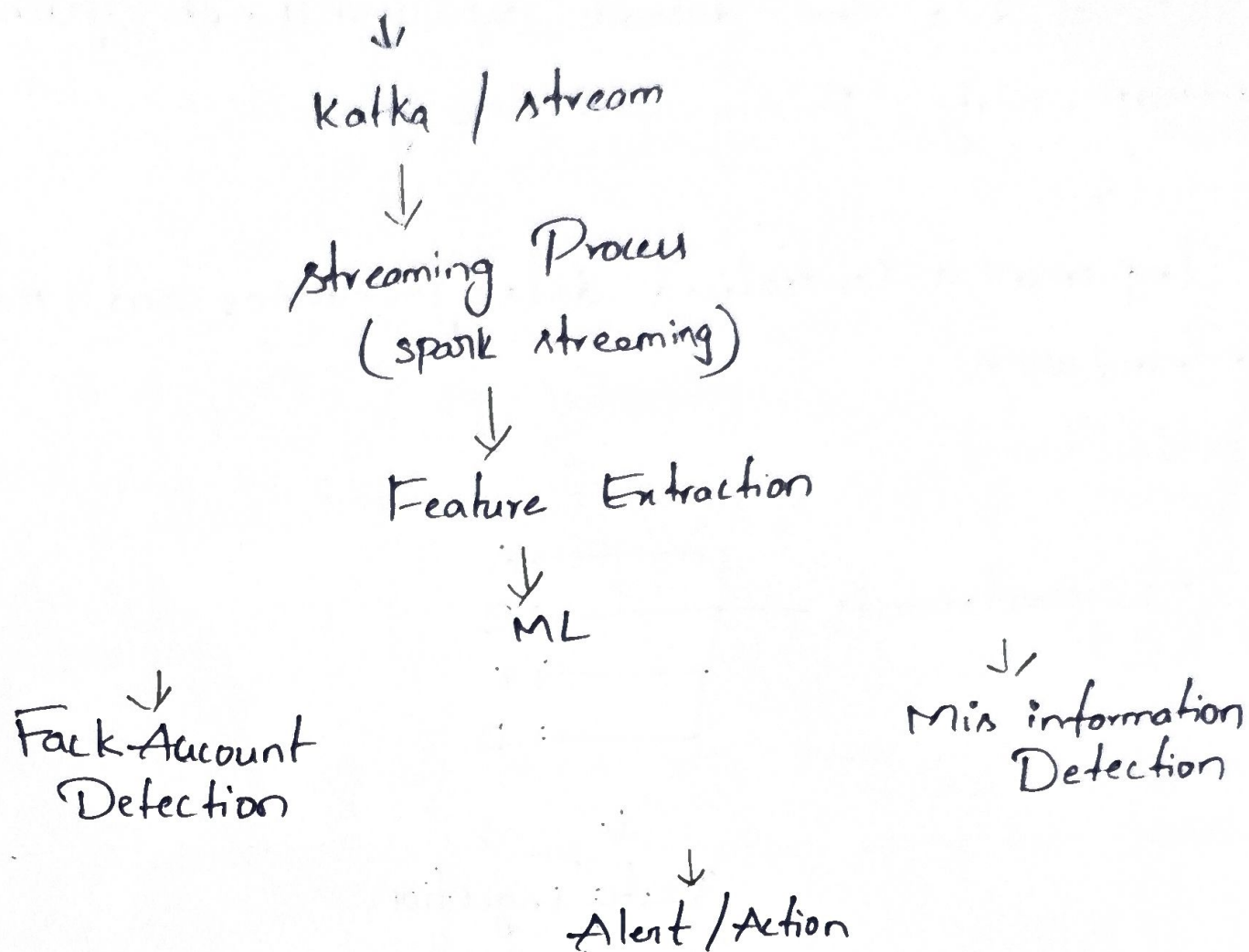
Security

Data quality

Authorized users

Architecture:

Posts / comments / likes / shares



Role of Machine Learning:

→ Analyze :- Account activity, Posting frequency, Network connections, Repeated content, Unusual behavior, Engagement patterns

Role of Stream Processing:

Stream Processing continuously analyzes incoming data instead of waiting for a large batch.

For example: New post → Process → Analyze → Risk source → Action.

Benefits:

1. Fast fraud detection

2. Scalable Processing

3. Continuous monitoring

4. Reduce manual efforts

Data governance ÷

1. Data Ownership → Identify who is responsible for each dataset.
2. Data quality → remove duplicate, incorrect, and incomplete
3. metadata → management maintain info - about dataset.
4. Access control → allow only Authorized users.
5. Data standards → use common formats so dept can exchange data.
6. Auditing → record data access and modifications.

Security ÷ Encryption, Authentication, Role-based access control,
Secure APIs, Access logging, Security audit.

Benefits: Data sharing b/w dept, interoperability, data quality etc.

4 Banking fraud Detection Problem ÷

A bank processing millions of transactions every minute and needs low-latency, high accuracy fraud detection using both historical and streaming data.

Big Data Pipeline

Bank transactions

Kafka/Stream

Realtime Processing

Feature extraction

Fraud Detection model

Pinkecore

Normal

Approve

Suspicious

Alert / Block

Historical data

Historical Transactions

Big Data storage

Machine Learning

Fraud Detection model

Real time system

Technologies :

1. Kafka : receives transaction streams.
2. Spark Streaming : Processes transactions
3. Big data storage : stores historical transactions
4. Machine Learning : identifies suspicious transactions
5. NoSQL / database → Provide fast access.

Algorithms :

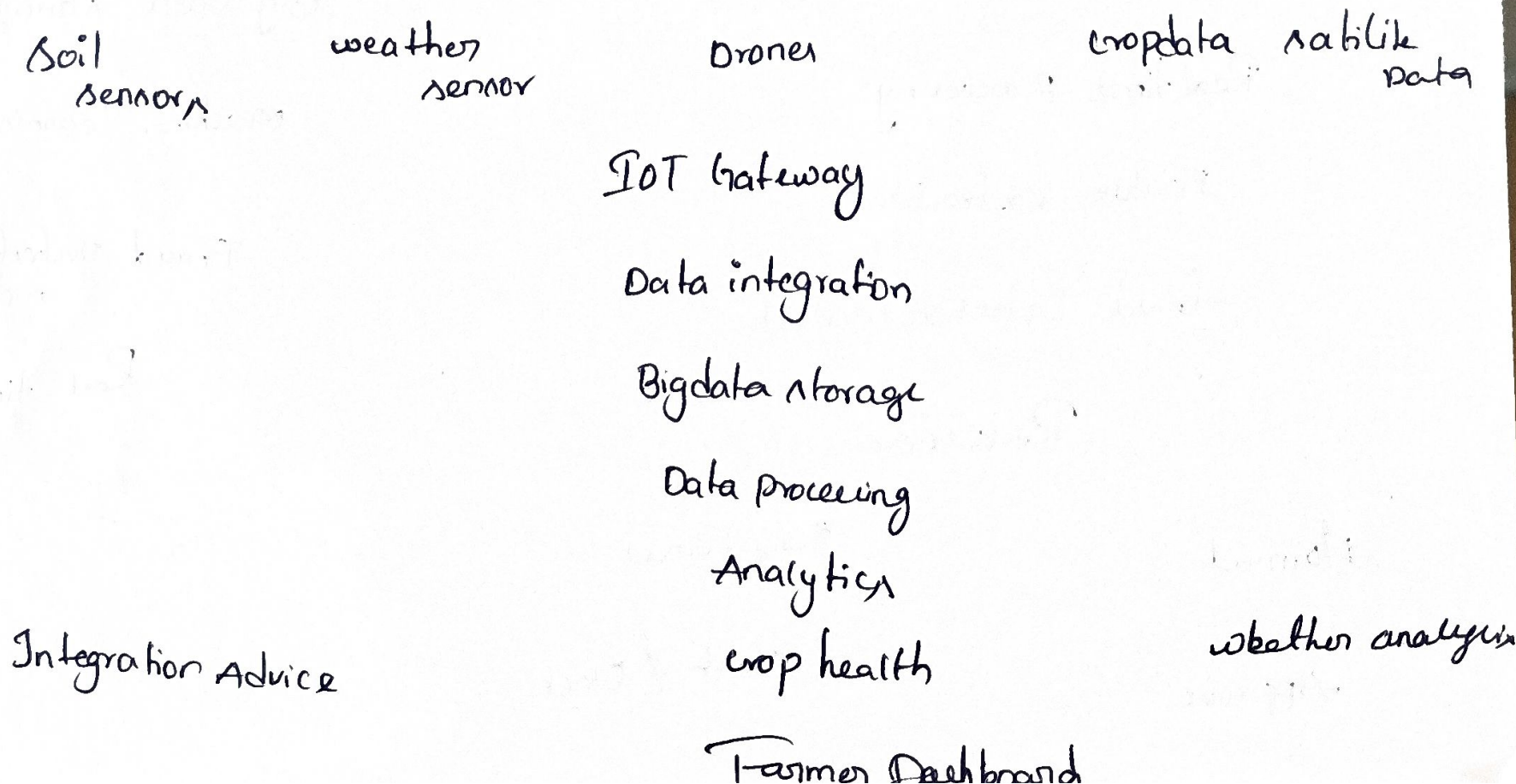
1. Logistic Regression, Decision tree, Random forest, Gradient Boosting, Anomaly Detection.

Benefits : Real time fraud Detection, Reduced financial loss, low response time, Accuracy high, monitoring early.

5. Agriculture Smart Farming Problem

Smart farming collects soil, weather, crop, sensor, and drone data from distributed sources.

Architecture :



Analytics :

1. Soil analysis
2. Irrigation Prediction
3. Crop health monitoring
4. Yield prediction
5. Resource optimization

Technologies :

1. IOT Sensors
2. Cloud computing
3. Bigdata storage
4. Streaming process
5. Machine Learning
6. Data Visualization dashboards

Benefits : Improved crop yield, Efficient water usage,
Early identification of crop problems, better decision making,
Reduced resource wastage